Lab 3
Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET)

Electronics and Photonics Post-undergraduate Second Degree Program

National Yang Ming Chiao Tung University

Outline

01 MOSFET Biasing + Switch

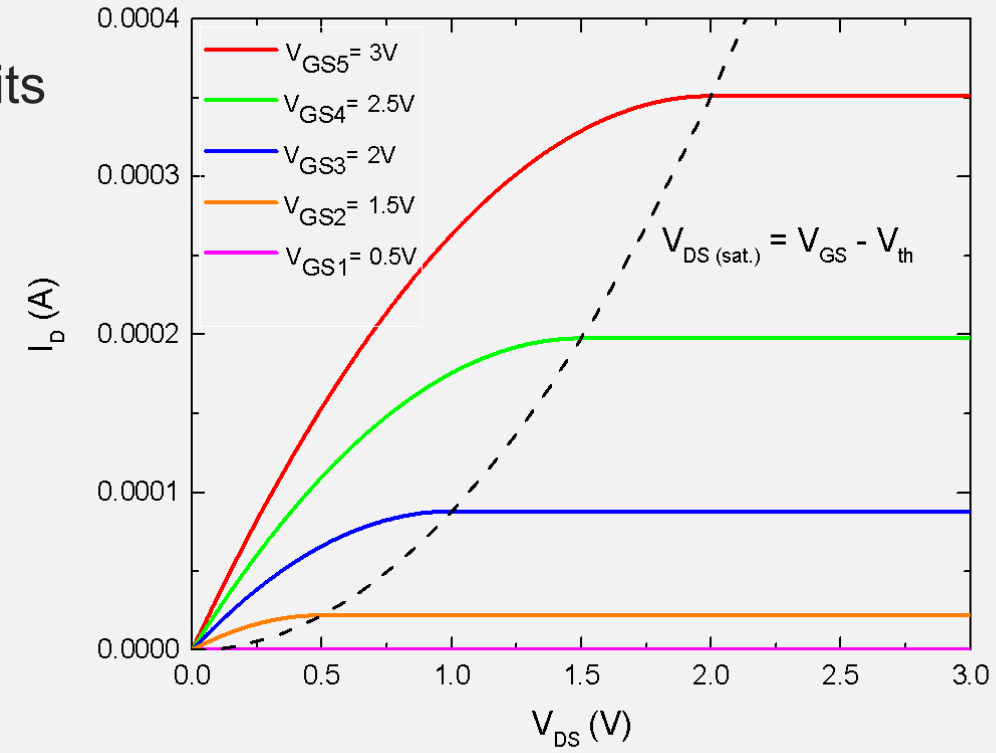
02 MOSFET as an Amplifier (Common-Source)

MOSFET Biasing + Switch

Objective

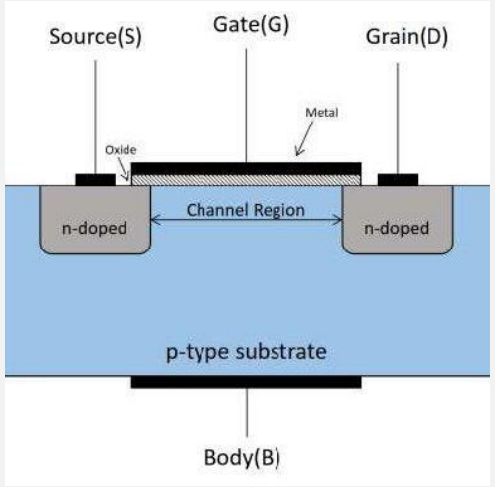
❿Build-up the concepts for MOSFET operation regions

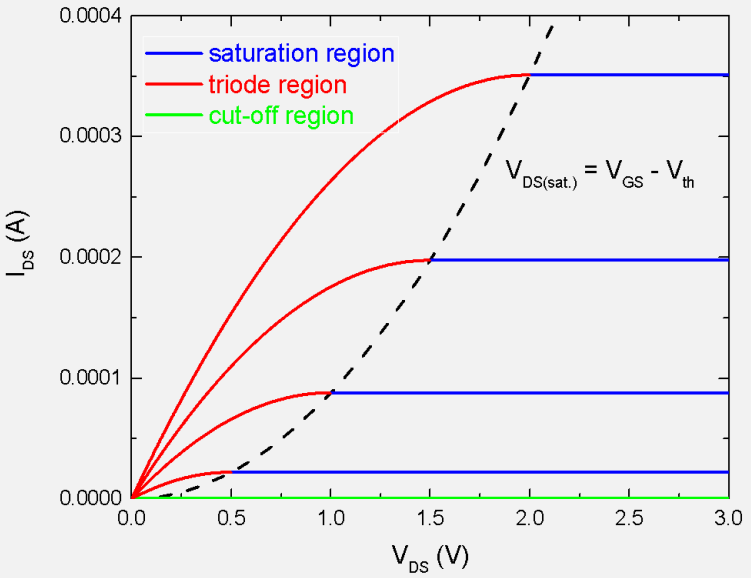
- ➢Saturation region
  - ✓Application: signal amplification in analog circuits
- ➢Cut-off region
  - ✓Application: turn-off state in digital circuit
- ➢Triode region
  - ✓Application: turn-on state in digital circuit



Theory overview

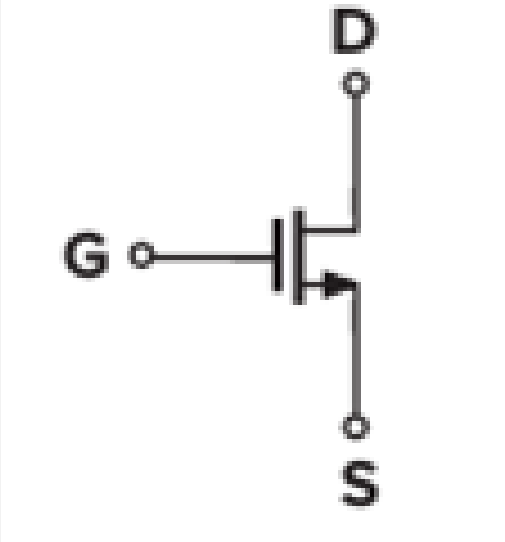
- Saturation region
  - $V_{GS} > V_{th}$
  - $V_{GD} < V_{th}$
- Cut-off region
  - $V_{GS} < V_{th}$
- Triode region
  - $V_{GS} > V_{th}$
  - $V_{GD} > V_{th}$





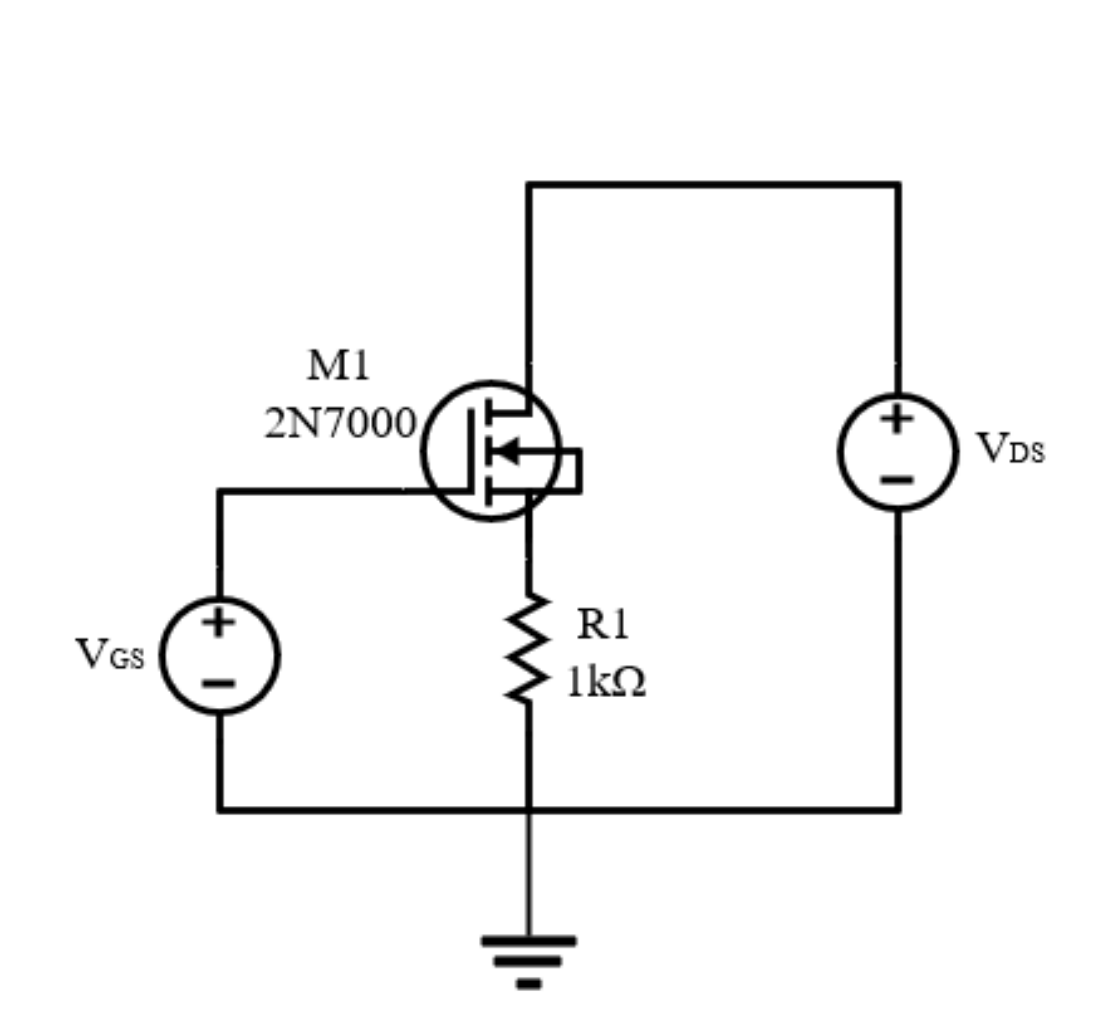
$$I_D = \frac{1}{2} * \mu * C_{OX} * \frac{W}{L} * (V_{GS} - V_{th})^2$$

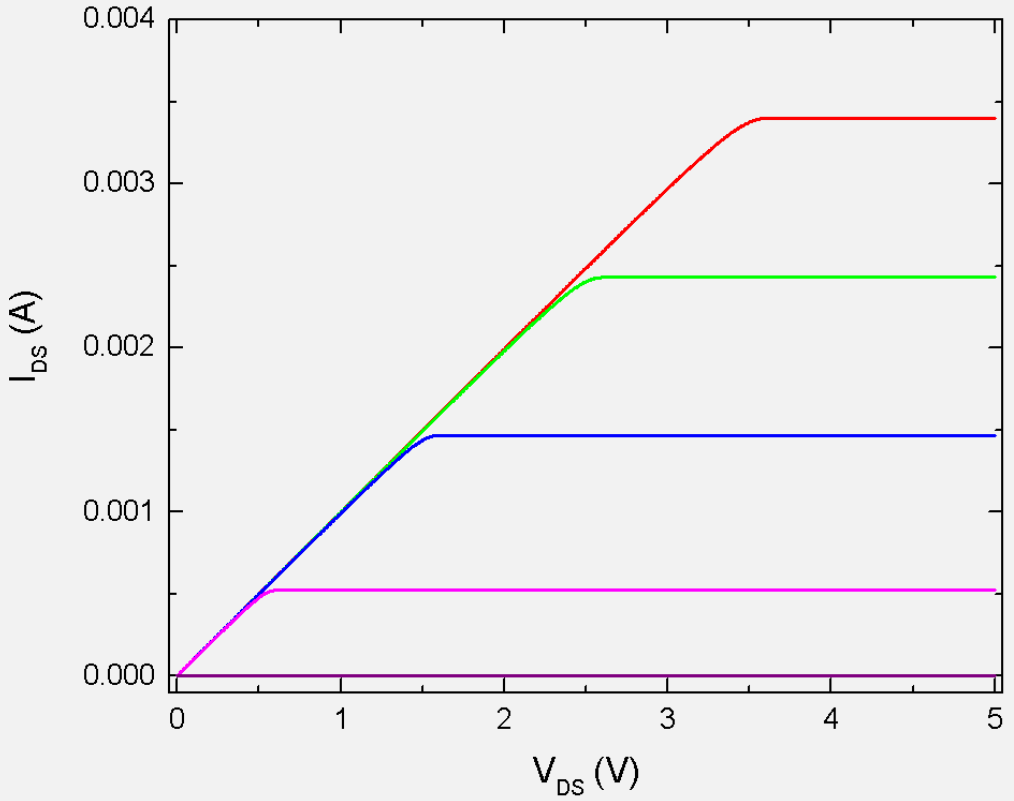
$$I_D = \frac{1}{2} * \mu * C_{OX} * \frac{W}{L} * [2 * (V_{GS} - V_{th}) * V_{DS} - V_{DS}^2]$$



Key point: V_{GS} ,V_{GD} and V_{th}

Circuit schematic (characteristic curve, I_D-V_{DS})





Requirement: characteristic curve

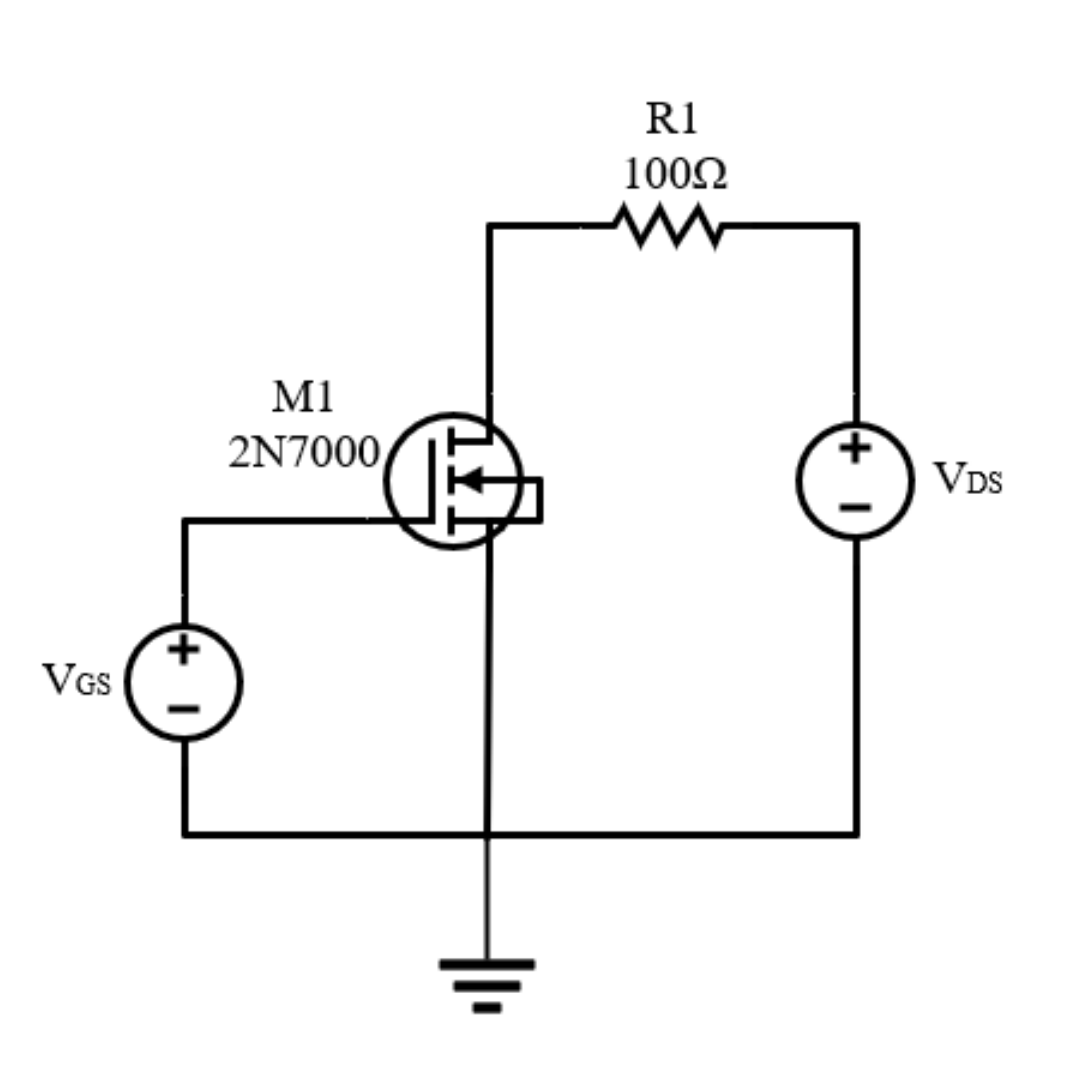
V_{GS} = 0 V	
V_{DS} (experimental setting)	I_D (measure)
1 V	A
2 V	A
V	A
V	A
V	A

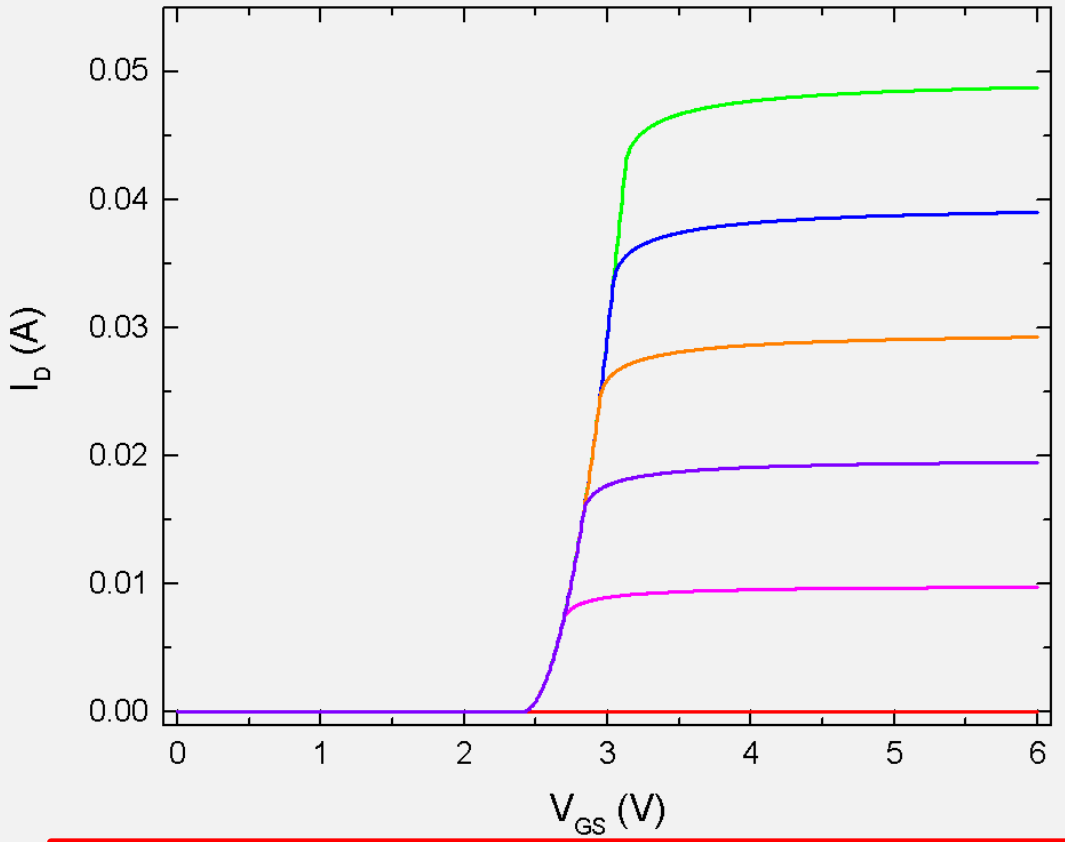
V_{GS} = 4 V	
V_{DS} (experimental setting)	I_D (measure)
1 V	A
2 V	A
V	A
V	A
V	A

V_{GS} = 2 V	
V_{DS} (experimental setting)	I_D (measure)
1 V	A
2 V	A
V	A
V	A
V	A

V_{GS} = 6 V	
V_{DS} (experimental setting)	I_D (measure)
1 V	A
2 V	A
V	A
V	A
V	A

Circuit schematic (transfer curve, I_D-V_{GS})

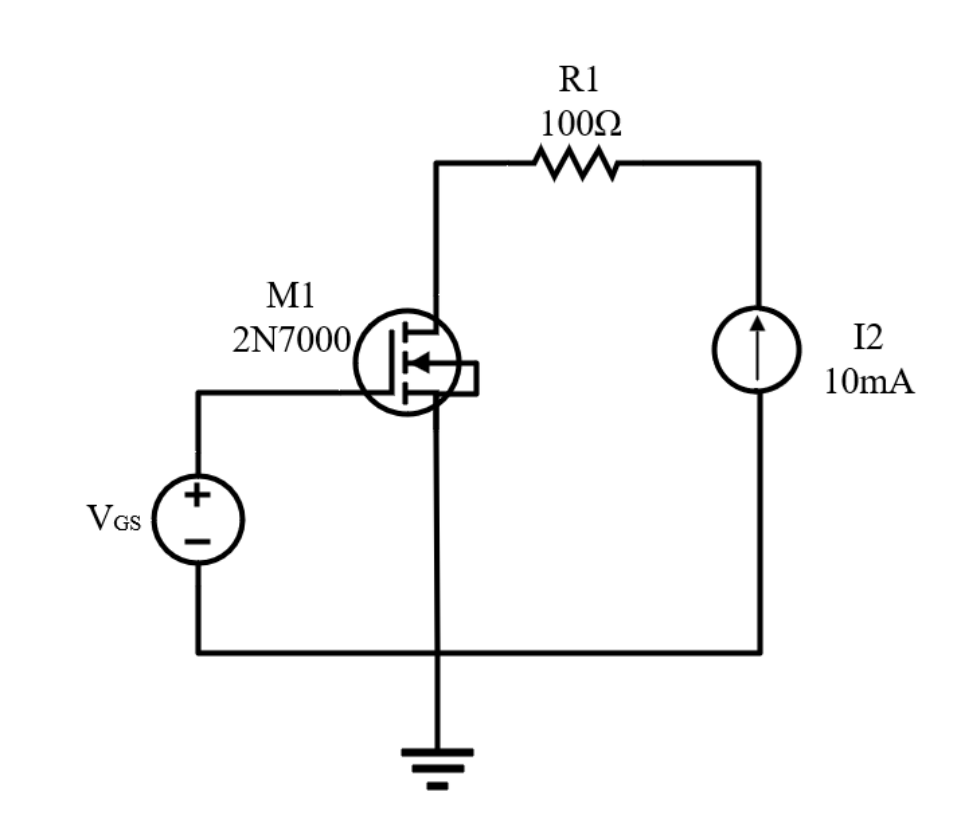


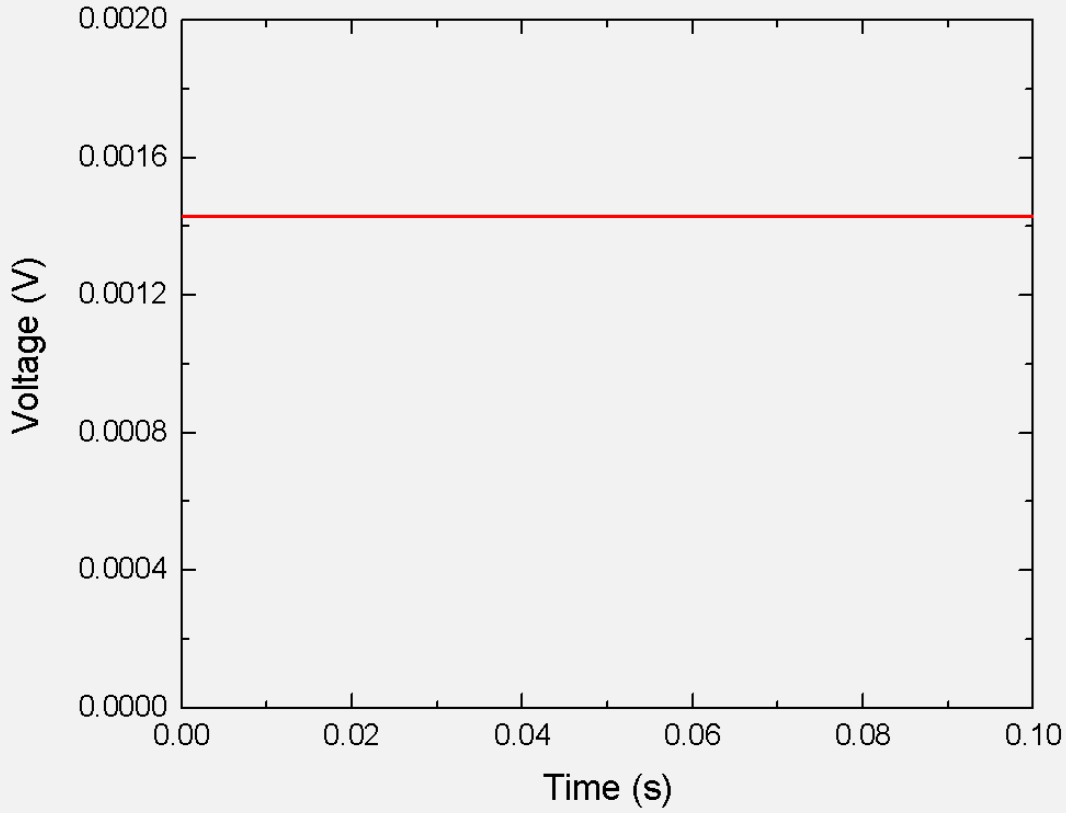


Requirement: transfer curve, threshold voltage

V_{DS} = 1 V	
V_{GS} (experimental setting)	I_D (measure)
0.5 V	A
1 V	A
V	A
V	A
V	A
V	A
V_{DS} = 3 V	
V_{GS} (experimental setting)	I_D (measure)
0.5 V	A
1 V	A
V	A
V	A
V	A
V	A
V_{DS} = 5 V	
V_{GS} (experimental setting)	I_D (measure)
0.5 V	A
1 V	A
V	A
V	A
V	A
V	A

Circuit schematic (turn-on resistance)

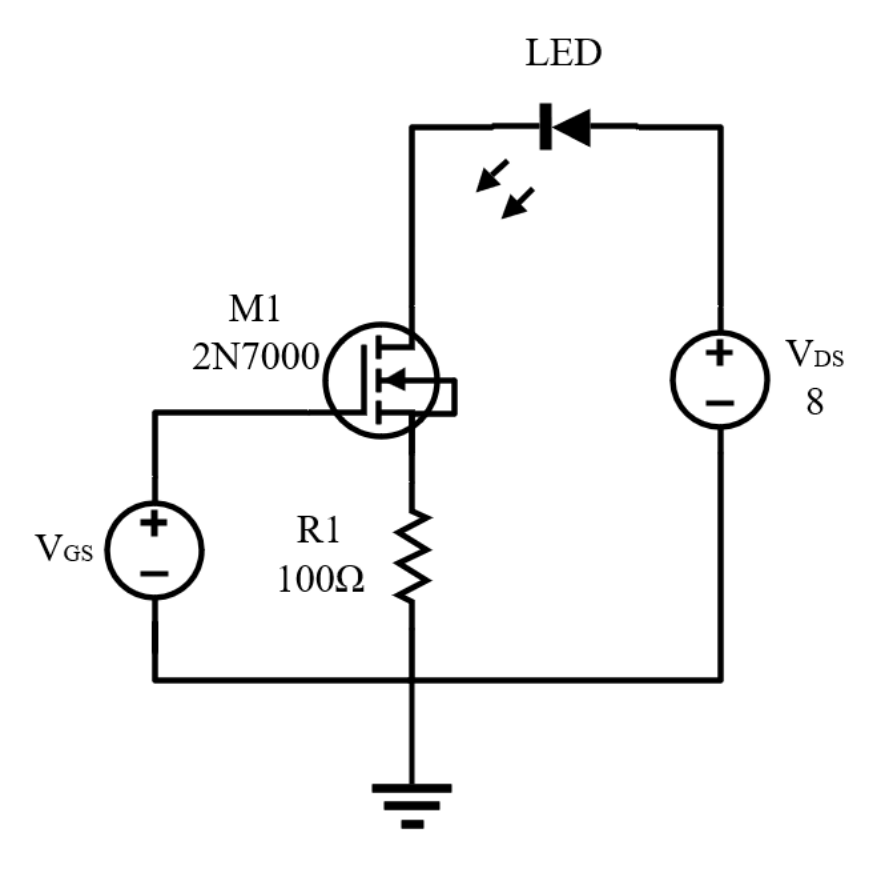


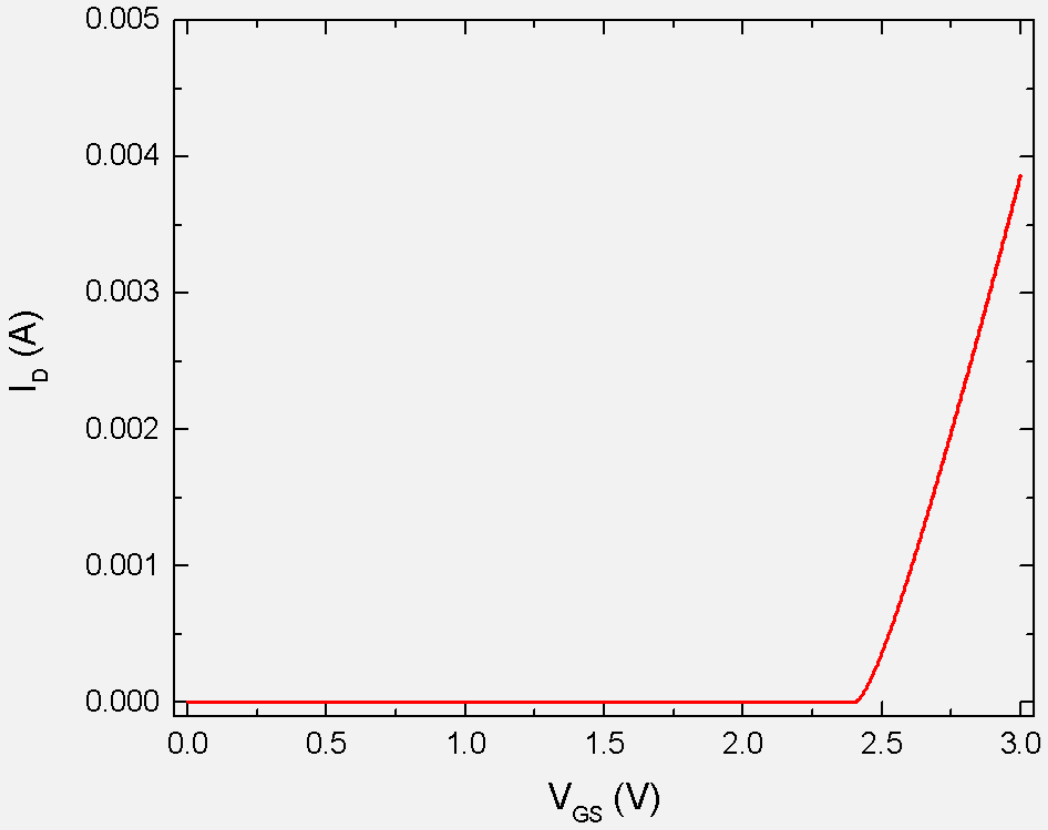


V_{GS} = 5 V and I_D = 10m A	
V_{DS} (measure)	R_{on} (calculation)
V	Ohm

Requirement: turn-on resistance

Circuit schematic (switch)

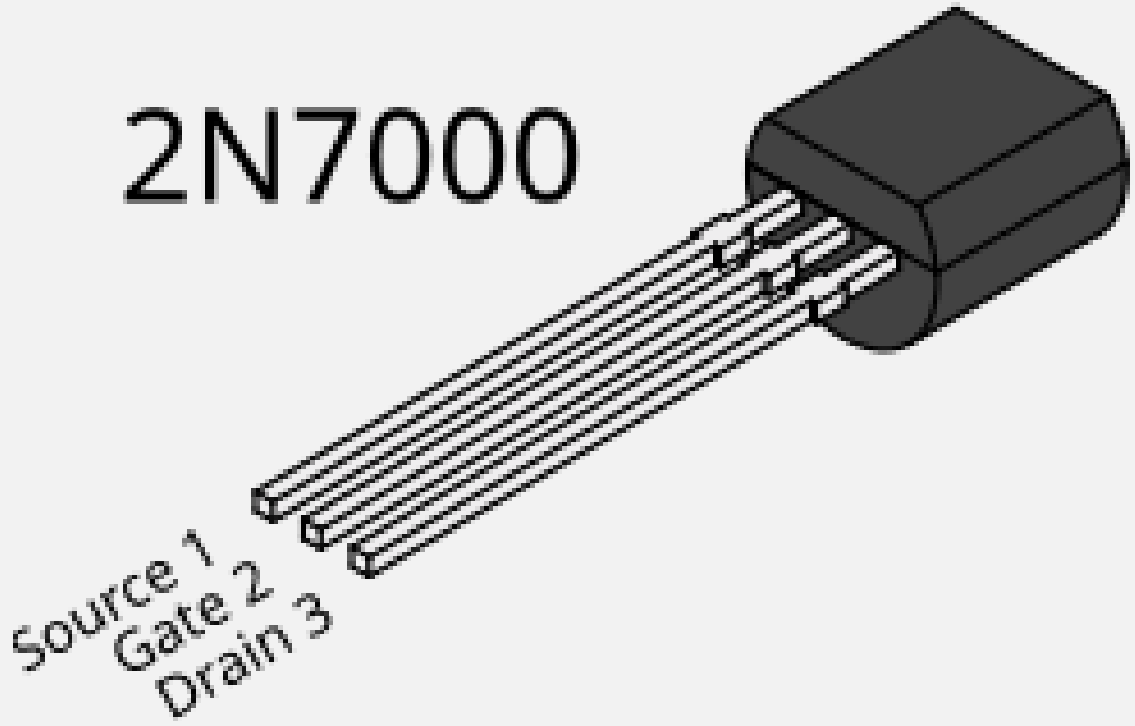




Requirement: cut-off and triode region observation, LED resistance

V_{DS} = 8 V		
V_{GS} (experimental setting)	I_D (measure)	LED performance
0.5 V	A	Dark / Bright
1 V	A	Dark / Bright
V	A	Dark / Bright
V	A	Dark / Bright
V	A	Dark / Bright
V	A	Dark / Bright

MOSFET Device (2N7000)



- N-type MOSFET
- Enhancement-mode MOSFET
- Low-power switching application

Reference: Wikipedia

Component list

Component	Specification	Number
MOSFET	2N7000	1
Resistor	100 Ω (1/4 W)	1
	1 kΩ (1/4 W)	1
DC Power Supply	GW Instek GPE-3323 (Kaohsiung) UTP-3305-II (Hsinchu)	1
Signal Source (Function Generator)	Keysight EDUX1052G (Kaohsiung) GW Instek AFG-2225 (Hsinchu)	1
Oscilloscope	Keysight EDUX1052G (Kaohsiung) OSD-1102 (Hsinchu)	1
Multimeter	GW Instek GDM-532 (Kaohsiung) DHA DMM-93B+ (Hsinchu)	1

Question

1. Using the characteristic curve obtained from Experiment 1, generate the corresponding transfer curve.

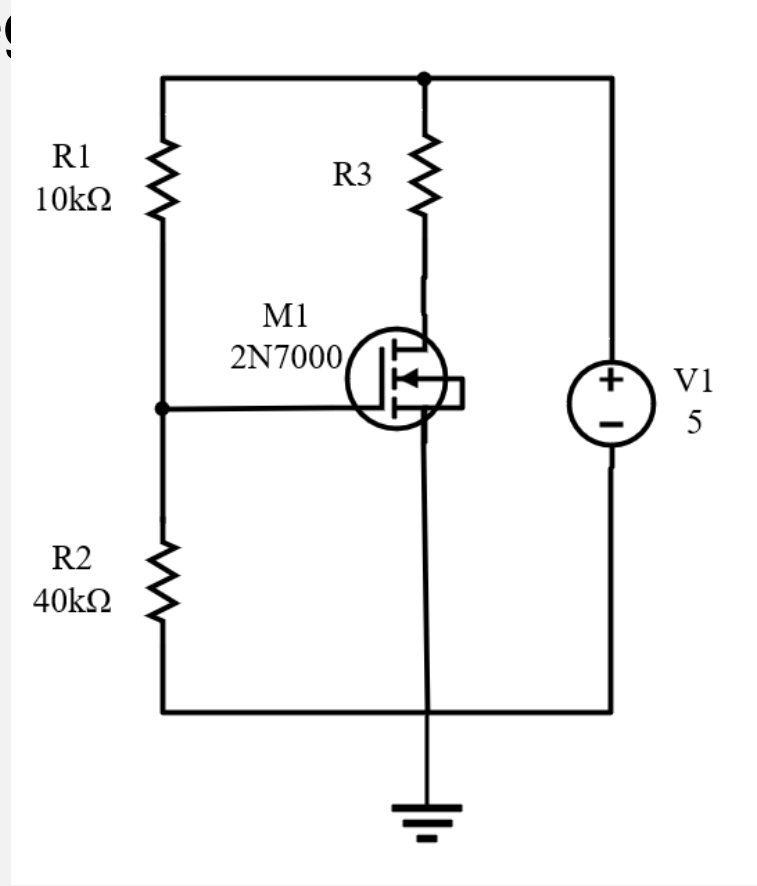
2. Using the transfer curve obtained from Experiment 2, explain three operation regions corresponding the curve.

3. In Experiment 3, with V_{GS} = 10V, what is the observed turn-on voltage of the MOSFET? Is it the same as previously measured? If not, explain the reason for any discrepancy.

4. If the resistor in Experiment 4 is replaced with a 200 Ω resistor, how does this affect the experiment compared to the original configuration?

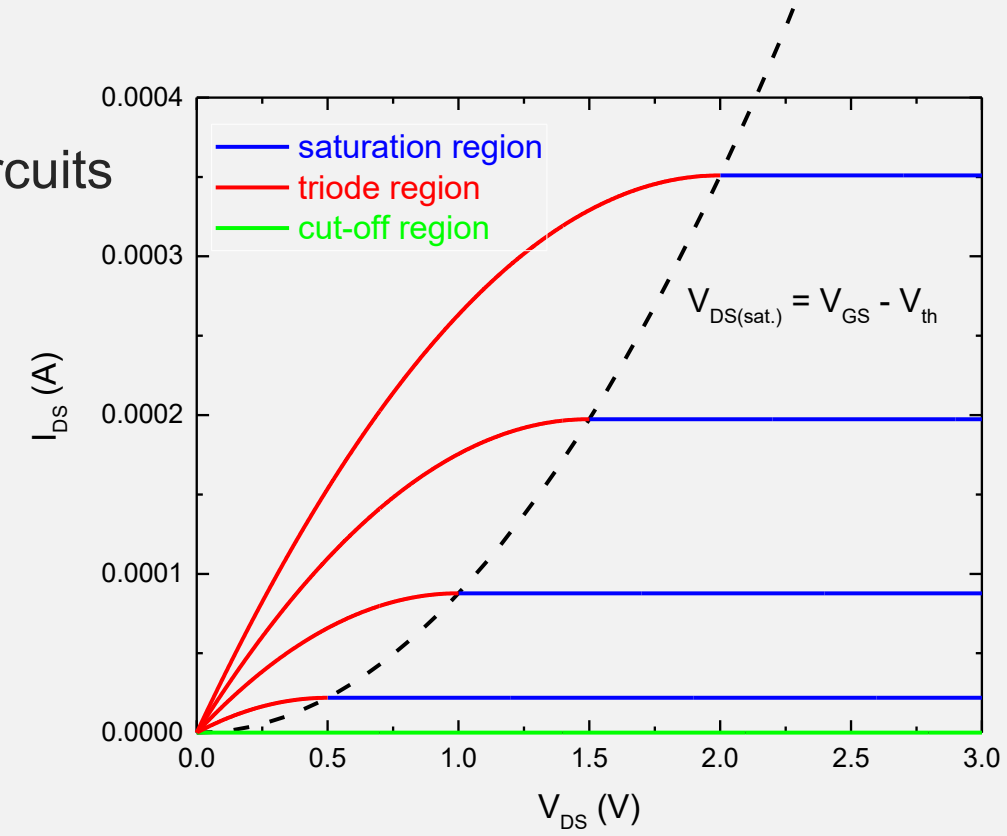
Bonus

1. By the threshold voltage in the experiment calculation, design the vibration region of R_3 value to make sure the MOSFET work in saturation re



Objective

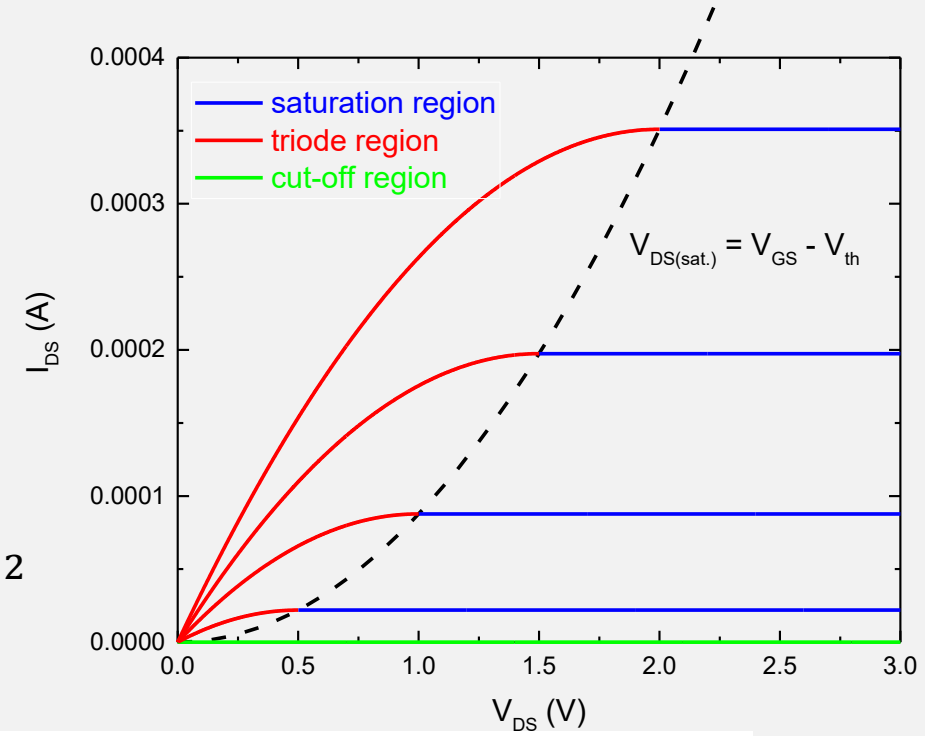
- Build-up the concept for MOSFET saturation region
  - Saturation region
    - ✓ Application: signal amplification in analog circuits
  - Amplification (gain)
  - Operation point
  - Impedance matching



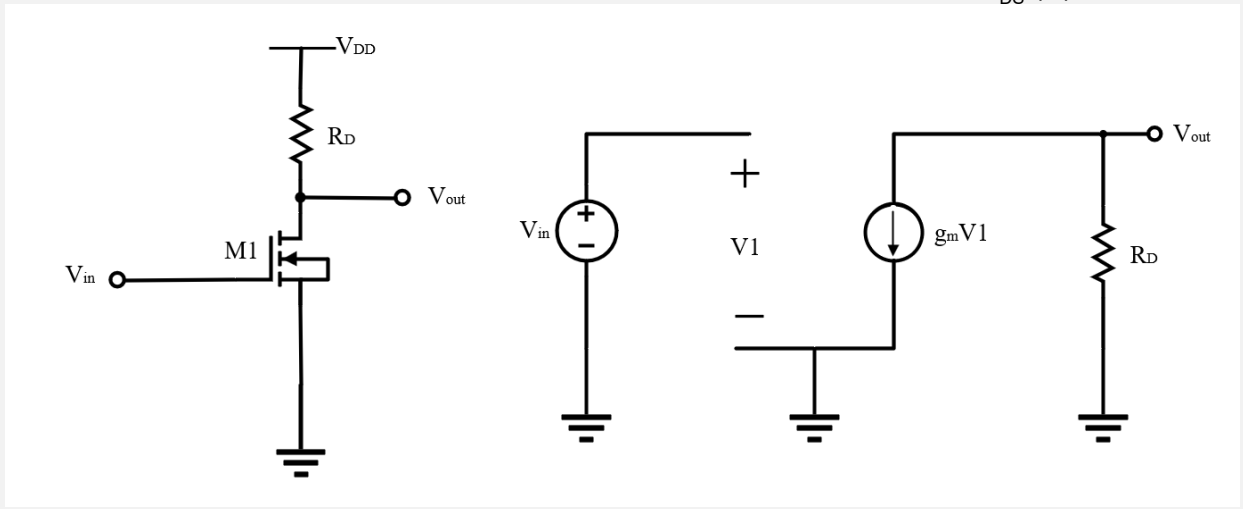
Theory overview

- Saturation region
  - $V_{GS} > V_{th}$
  - $V_{GD} < V_{th}$

$$I_D = \frac{1}{2} * \mu * C_{OX} * \frac{W}{L} * (V_{GS} - V_{th})^2$$

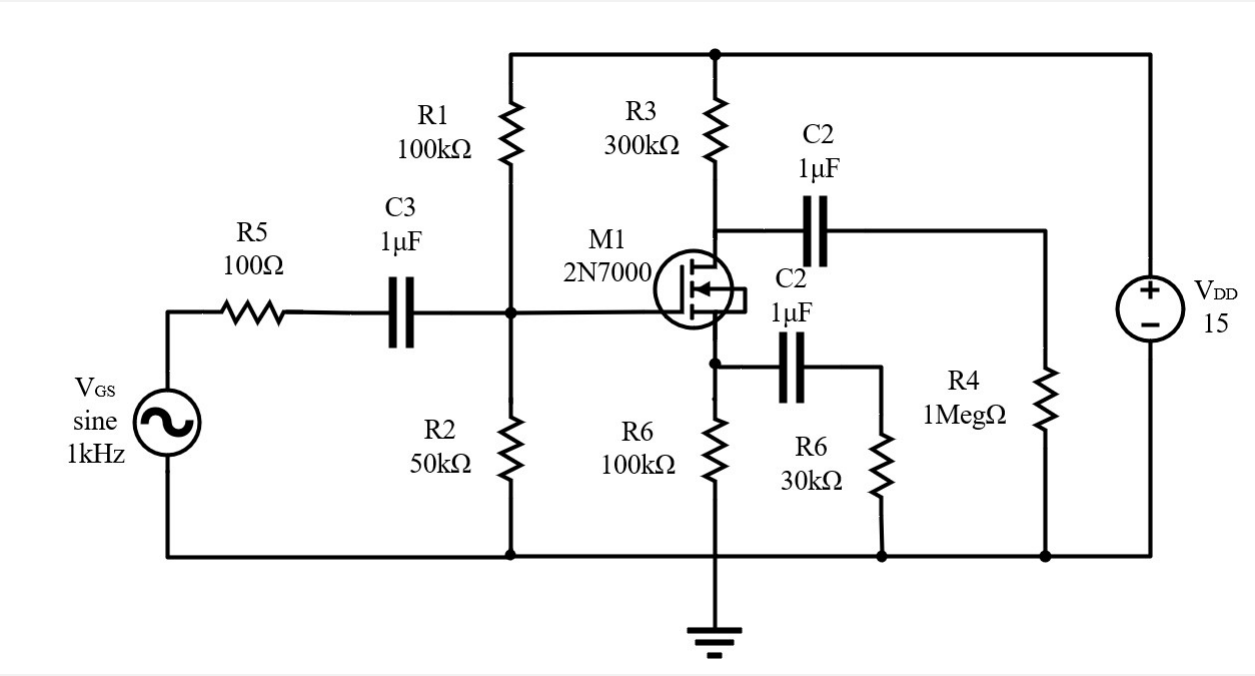


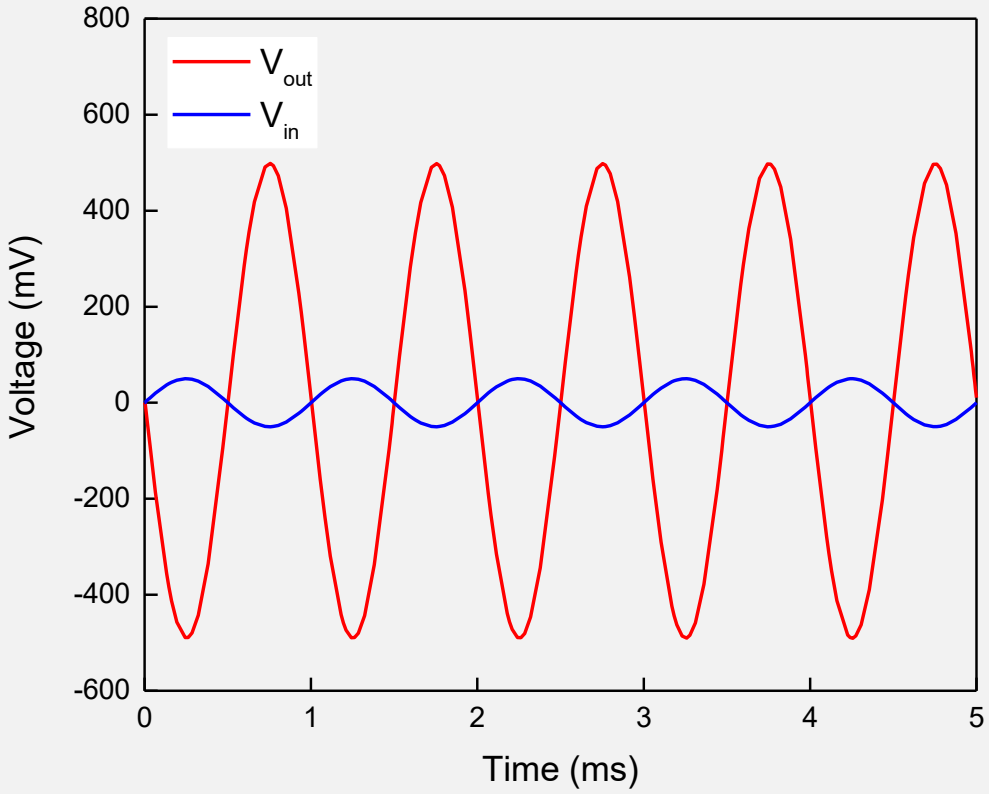
- Voltage gain
  - $g_m = \frac{dI_D}{dV_{GS}} = \frac{2*I_D}{V_{GS}-V_{th}}$
  - $A_V = \frac{V_{out}}{V_{in}}$
  - $A_V = -g_m * R_D$



Key point: V_{GS} , V_{GD} , V_{th} and g_m

Circuit schematic





Requirement: operation point and voltage gain

$V_{DD} = 15$ V					
V_{GS} (theory)	V_{GS} (measure)	V_{DS} (theory)	V_{DS} (measure)	Operation region	g_m (theory) (Hint: I_D measurement)
V	V	V	V		Ohm^{-1}

$V_{GS} = 0.05 \times \sin(2 \times \pi \times 1000 \times t)$ and $V_{DD} = 15$ V				
V_{out} (Max.) (theory)	V_{out} (measure)	A_v (Max.) (theory)	A_v (calculation)	g_m (calculation)
V	V			Ohm^{-1}

Component list

Component	Specification	Number
MOSFET	2N7000	1
Resistor	100 Ω (1/4 W)	1
	50 kΩ (1/4 W)	1
	50 kΩ (1/4 W)	1
	100 kΩ (1/4 W)	2
	300 kΩ (1/4 W)	1
	1 MΩ (1/4 W)	1
Capacitor	1 μF	3
DC Power Supply	GW Instek GPE-3323 (Kaohsiung) UTP-3305-II (Hsinchu)	1
Signal Source (Function Generator)	Keysight EDUX1052G (Kaohsiung) GW Instek AFG-2225 (Hsinchu)	1
Oscilloscope	Keysight EDUX1052G (Kaohsiung) OSD-1102 (Hsinchu)	1
Multimeter	GW Instek GDM-532 (Kaohsiung) DHA DMM-93B+ (Hsinchu)	1

Question

1. What is the function of $C_1 \& C_2$
2. Derive the equation of voltage gain for the experiment circuit configuration
3. Calculate the voltage gain when a capacitor is connected in parallel with R_6.
4. If the R_3 in Experiment is replaced with a 1 MΩ resistor, how does this affect the experiment compared to the original configuration?
5. If the R_1 in Experiment is replaced with a 1 MΩ resistor, how does this affect the experiment compared to the original configuration?

Bonus

1. Design the circuit according to the configuration shown below and fulfill the requirements

    1) The voltage gain must be greater than 10

    2) The output signal must be 180 degrees out of phase with the input signal

